CIVIL ENGINEERING					
IV B.Tech – I Semester					
Course Code	**GROUND IMPROVEMENT TECHNIQUES** **Professional Elective-V**	L	T	P	C
23A01703a		3	0	0	3

Course Objectives:

The objectives of this course are to make the student to:

1. **Understand** various dewatering methods, including sumps, well points, and electro-osmosis, for effective groundwater control.
2. **Analyze** the properties and applications of grouts, grouting techniques, and post-grouting tests for soil and rock stabilization.
3. **Evaluate** different densification techniques for granular and cohesive soils, such as vibro-compaction, preloading, and thermal methods.
4. **Apply** stabilization techniques, including mechanical, chemical, and bituminous stabilization, to improve soil properties.
5. **Assess** the design principles of reinforced earth walls and the roleofgeosyntheticsin soil improvement and slope stability. (Bloom's Level: **Evaluate - L5**

Course Outcomes (COs):

After successful completion of this course, students will be able to:

1. **Explain** the methods of dewatering and grouting and their importancein foundation engineering.
2. **Analyze** densification techniques for granular and cohesive soils to enhance soil strength.
3. **Evaluate** the effectiveness of soil stabilization methods for different ground conditions.
4. **Apply** reinforced earth principles andgeosynthetics for soil retention and foundation stability.
5. **Assess** expansive soil problems, their identification methods, and suitable foundation techniques like under-reamed piles.

CO – PO Articulation Matrix

Course Outcomes	**PO 1**	**PO 2**	**PO 3**	**PO 4**	**PO 5**	**PO 6**	**PO 7**	**PO 8**	**PO 9**	**PO 10**	**PO 11**	**PO 12**	**PSO1**	**PSO2**
CO -1	3	2	2	2	1	-	-	-	-	2	-	1	2	2
CO -2	3	3	2	2	2	-	-	-	-	2	-	2	2	2
CO -3	3	3	3	2	2	-	-	-	-	2	-	2	3	2
CO -4	3	2	3	3	3	-	-	-	-	2	1	2	3	3
CO -5	3	3	3	3	3	-	-	-	-	2	1	2	3	3

UNIT – I

EXPANSIVE SOILS: Problems of Expansive Soils – Tests for Identification – Methods of Determination of Swell Pressure. Improvement of Expansive Soils – Foundation Techniquesin Expansive Soils – Under Reamed Piles.

UNIT – II

DEWATERING: Methods of De-Watering- Sumps and Interceptor Ditches- Single, Multi Stage Well Points - Vacuum Well Points- Horizontal Wells-Foundation Drains-Blanket Drains

- Criteria for Selection of Fill Material Around Drains –Electro-Osmosis .

GROUTING: Objectives of Grouting- Grouts and Their Properties- Grouting Methods- Ascending, Descending and Stage Grouting- Hydraulic Fracturing in Soils and Rocks- Post Grout Test

UNIT – III

DENSIFICATION METHODS in GRANULAR SOILS:-

In – Situ Densification Methodsin Granular Soils:– Vibration At The Ground Surface, Impact At The Ground Surface, Vibration At Depth, Impact At Depth.

DENSIFICATION METHODS in COHESIVE SOILS:–

In – Situ Densification Methods in Cohesive Soils:– Preloading Or Dewatering, Vertical Drains – Sand Drains, Sand Wick Geodrains – Stone and Lime Columns – Thermal Methods

UNIT – IV

STABILISATION: Methods of Stabilization-Mechanical-Cement- Lime-Bituminous-Chemical Stabilization With Calcium Chloride, Sodium Silicate and Gypsum

UNIT – V

REINFORCED EARTH: Principles – Components of Reinforced Earth – Factors Governing Design of Reinforced Earth Walls – Design Principles of Reinforced Earth Walls.

GEOSYNTHETICS: Geotextiles- Types, Functions and Applications – Geogrids and Geo Membranes – Functions and Applications

TEXT BOOKS:

1. Engineering Principles of Ground Modification, Haussmann M.R. , McGraw-Hill International Edition(1990).
2. Ground Improvement Techniques, Dr.P.Purushotham Raj. Laxmi Publications, New Delhi University science press, New Delhi 2nd edition 2016

REFRENCE BOOKS:

1. Ground Improvement, Moseley M.P. Blackie Academic and Professional, Boca Taton, Florida, USA(1993).
2. Ground Improvement Techniques, NiharRanajanPatraVikas Publications, New Delhi
3. Ground Control and Improvement, Xanthakos P.P, Abramson, L.W andBrucwe, D.A (1994) John Wiley and Sons, New York, USA.

Online Learning Resources:

https://archive.nptel.ac.in/courses/105/108/105108075/
https://archive.nptel.ac.in/courses/105/105/105105210/